CASE STUDY REPORT

E-Waste Management

Case Studies for TE Students

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Introduction

Electronic waste (e-waste) is one of the fastest-growing waste streams in the world. It contains valuable metals such as copper, gold and silver, but also hazardous substances such as lead, mercury and cadmium. Improper handling harms human health and the environment, while proper recycling recovers resources and reduces mining pressure.

This report studies ten case studies covering international best practices (Japan and Switzerland), Indian policy (E-Waste Management Rules 2022 and EPR), Indian cities and states (Bengaluru, Mumbai, Kerala, Pune, Dharavi), and a corporate example (Dell). Each case study presents the background, key features, analysis and the lessons relevant to India.

Case Study 1: Why Japan Recycles More Than Most Countries: Lessons for India

Background

Japan is a resource-poor, densely populated island nation with limited landfill space. This scarcity pushed it to treat discarded products as a source of secondary raw materials. Its legal framework is built around the Basic Act for Establishing a Sound Material-Cycle Society, supported by product-specific laws.

Key Features

- Home Appliance Recycling Law (in force 2001): covers air conditioners, TVs, refrigerators/freezers and washing machines. Consumers pay a recycling fee at the time of disposal, retailers take back old units, and manufacturers must recycle them to prescribed rates.
- Small Home Appliance Recycling Law (2013): promotes collection of small devices such as phones, cameras and laptops for metal recovery, mainly through municipalities.
- Tokyo 2020 Olympic medals were made entirely from metals recovered from donated phones and small electronics, a nationwide public-awareness campaign.

Analysis

Japan's success rests on three pillars: clear legal duties for every stakeholder, a culture of source segregation, and strong industrial recycling capability. Citizens are accustomed to sorting waste into many categories, and manufacturers design products with later dismantling in mind.

Lessons for India

India can adopt retailer take-back, visible consumer-fee or deposit mechanisms, public awareness campaigns, and design-for-recycling norms. Cultural change in segregation takes time and must be supported by convenient collection points.

Case Study 2: Switzerland's E-Waste Success Story

Background

Switzerland was one of the first countries in the world to organise e-waste recycling, starting with refrigerators in the early 1990s. The system is built on producer responsibility and is run by industry-managed organisations rather than by the government directly.

Key Features

- Ordinance on the Return, Taking Back and Disposal of Electrical and Electronic Equipment (ORDEA, 1998) legally obliges retailers, importers and manufacturers to take back old equipment free of charge.
- Advance Recycling Fee (ARF): a small fee included in the purchase price funds collection and recycling. Organisations such as SENS eRecycling, SWICO and SLRS administer the funds.
- A dense network of retail take-back points and municipal collection sites means consumers rarely need to travel far. Switzerland collects roughly 15 kg of e-waste per person per year, among the highest rates globally.

Analysis

The model succeeds because costs are paid up-front, so no one needs to be chased at the end of the product's life. Recyclers are audited and must meet technical standards, which prevents illegal export and unsafe processing.

Lessons for India

An advance fee with transparent financial management is a strong alternative to certificate-based EPR. India could pilot a similar fee for high-volume categories and publish collection and recycling data openly.

Case Study 3: India's E-Waste Management Rules 2022

Background

India is among the largest generators of e-waste in the world, after China and the USA. The E-Waste (Management) Rules, 2022 came into force on 1 April 2023, replacing the 2016 Rules, to formalise the sector and strengthen Extended Producer Responsibility.

Key Features

- Applies to manufacturers, producers, refurbishers and recyclers; covers 106 types of electrical and electronic equipment listed in Schedules.
- Mandatory online registration on the Central Pollution Control Board (CPCB) portal for producers, refurbishers and recyclers.
- EPR certificates are generated by recyclers and refurbishers and purchased by producers to meet annual targets; a pricing framework for certificates and environmental compensation for non-compliance are included.
- Restrictions on hazardous substances (lead, mercury, cadmium, hexavalent chromium, PBB and PBDE) continue as in the earlier rules.

Analysis

The 2022 Rules move from a purely target-based system to a trackable, market-linked one. The digital portal improves transparency, and the inclusion of refurbishers promotes reuse. However, enforcement capacity at state level and the role of the informal sector remain challenges.

Lessons for India

Clear rules are necessary but not sufficient: monitoring, audits, consumer awareness and integration of informal workers decide actual outcomes.

Case Study 4: India's Extended Producer Responsibility (EPR): Success or Paper Compliance?

Background

EPR makes producers financially and physically responsible for the end-of-life management of their products. India introduced EPR for e-waste in 2011 and reformed it in 2016 and 2022. The key question is whether it improves real recycling or merely produces paperwork.

Key Features

- Under the 2022 Rules, producers meet targets by purchasing EPR certificates rather than necessarily collecting waste themselves.
- Concerns have been raised about certificates being generated without matching physical processing, and about weak verification of recyclers' records.
- Producers' collection targets are based on past sales, which is difficult to verify for imported or unregistered goods.
- The informal sector, which handles a large share of e-waste, is largely outside the certificate system.

Analysis

EPR has increased the number of registered recyclers and brought producers into the system, which is a real success. At the same time, the certificate-trading design can reward paperwork over real recovery if audits are weak. Regulators have acted against fraudulent certificates, showing that the system is being tightened.

Lessons for India

EPR will deliver results only when it is combined with random physical audits, traceability of material flow, fair certificate pricing, and formal linkages with collectors from the informal sector.

Case Study 5: Bengaluru: India's E-Waste Capital?

Background

Bengaluru, as India's IT and electronics hub, generates large volumes of e-waste from corporate offices, households and start-ups. Karnataka is also home to a significant number of authorised recyclers, which is why the city is often described as an e-waste capital.

Key Features

- Large corporate and institutional (bulk) consumers generate obsolete computers, servers and peripherals.
- Authorised recyclers such as E-Parisaraa, one of the early formal recyclers in the state, process e-waste using environmentally sound methods.
- The Karnataka State Pollution Control Board (KSPCB) authorises and monitors recyclers; the city's civic body runs collection drives and drop-off points.
- An active informal sector operates alongside formal recyclers, especially for household devices.

Analysis

Bengaluru shows how a technology hub both creates the e-waste problem and the talent and entrepreneurship to solve it. The main weakness is leakage: household e-waste still frequently goes to scrap dealers instead of authorised channels.

Lessons for India

Cities with high e-waste generation should have convenient collection points, bulk-consumer tie-ups and public awareness, supported by strong monitoring by the pollution control board.

Case Study 6: Mumbai's E-Waste Ecosystem

Background

Mumbai, the financial capital, and Maharashtra as a state are among the largest generators of e-waste in India. The city has a mixed ecosystem of authorised recyclers, collection agencies, scrap dealers and informal dismantlers.

Key Features

- Authorised recyclers in and around Mumbai, such as Ecoreco, operate under Maharashtra Pollution Control Board authorisation.
- Informal markets and scrap dealers collect, repair and dismantle devices, supplying much of the secondary market.
- Municipal and NGO-led collection drives, and producer take-back programmes, supplement the formal system.
- High land cost and limited space make setting up large processing facilities within the city difficult.

Analysis

The strength of Mumbai's ecosystem is its volume and the availability of repair and resale skills. The weakness is fragmentation: formal and informal actors rarely work together, and valuable material is often recovered by crude methods such as open burning or acid leaching.

Lessons for India

Bridging formal recyclers and informal collectors through registration, training and fair pricing can raise collection rates while improving worker safety and environmental performance.

Case Study 7: Kerala's Clean Kerala Company

Background

Clean Kerala Company Limited (CKCL) was set up by the Government of Kerala, under the Local Self Government Department, to manage non-biodegradable waste in a scientific manner. It works with local bodies and the Haritha Karma Sena, a network of women-led green workers.

Key Features

- Door-to-door collection of non-biodegradable waste by Haritha Karma Sena members, who earn from user fees.
- Material Collection Facilities (MCFs) at local-body level where waste is sorted and aggregated before it is sent to recyclers.
- Periodic e-waste collection drives in partnership with authorised recyclers, which gather discarded electronics from households and institutions.
- Strong participation of local self-government institutions and community groups.

Analysis

Kerala's decentralised, community-based model has high social acceptance and creates local green jobs. Challenges include the cost of logistics, fluctuating prices of recyclables and the need for regular funding and monitoring.

Lessons for India

A state-owned or state-backed company with local-body partnerships can create a sustainable collection backbone that formal recyclers and EPR producers can plug into.

Case Study 8: Pune's SWaCH Model

Background

SWaCH (Solid Waste Collection and Handling) is a wholly owned cooperative of self-employed waste pickers and other urban poor in Pune. Established in 2007 in partnership with the Pune Municipal Corporation, it grew out of the work of the waste pickers' union, KKPKP.

Key Features

- Members provide door-to-door collection of segregated waste and are paid user fees directly by households, in addition to income from selling recyclables.
- Waste pickers are given identity cards and formal recognition, improving their social standing and access to services.
- The cooperative also collects dry waste, including small electronics, and routes them to appropriate recyclers.
- Citizens are encouraged to segregate waste at source, which raises the quality of recoverable material.

Analysis

SWaCH is widely cited as a model of inclusive waste management: it improves collection efficiency, reduces the load on landfills and provides dignified livelihoods. Its sustainability depends on continued municipal support and fair user fees.

Lessons for India

Integrating informal workers into the formal system, rather than displacing them, is both socially just and operationally effective, and is directly relevant to e-waste collection under EPR.

Case Study 9: Dharavi Recycling Ecosystem

Background

Dharavi in Mumbai is one of the largest informal settlements in Asia and houses a dense cluster of recycling and manufacturing micro-enterprises. Estimates of its recycling industry commonly cite thousands of small units and a very large workforce, though exact numbers vary by source.

Key Features

- Plastic, paper, metal, textile, glass and some electronic waste are collected, sorted, cleaned and reprocessed within the settlement.
- The system supports livelihoods for a large number of low-income workers and diverts significant waste from landfills at almost no cost to the city.
- Working conditions are often unsafe, with exposure to toxic fumes, chemicals and sharp materials, and with little protective equipment.
- Redevelopment plans for Dharavi raise questions about the future of these businesses and workers.

Analysis

Dharavi demonstrates the economic value and efficiency of informal recycling, but also its environmental and health costs. Informal methods for e-waste, such as burning wires or using acids, can release hazardous substances.

Lessons for India

Upgrading informal recyclers through training, safe equipment, shared common facilities and legal recognition can combine the strengths of the informal sector with environmental safety.

Case Study 10: Dell's Circular Economy Strategy

Background

Dell Technologies is a global IT hardware company that has made circular economy principles part of its product and business strategy. It focuses on using recycled materials, extending product life and recovering used products at end of life.

Key Features

- Take-back and recycling programmes for customers, including Dell Reconnect, a partnership with Goodwill that began in 2004, along with asset resale and recovery services for businesses.
- Closed-loop recycling: plastics and metals recovered from used electronics are used again in new products; Dell has also used gold recovered from old motherboards in new products.
- Use of recycled and renewable materials in products and sustainable packaging, with public goals for 2030 to increase recycled or renewable content and make all packaging recyclable or compostable.
- Design for longevity, repairability and modularity to lengthen product life.

Analysis

Dell's approach shows how a producer can turn EPR obligations into a business opportunity by reducing raw material cost and improving brand value. Challenges include recovering enough used products in a consistent quality and the cost of closed-loop processing.

Lessons for India

Indian producers can go beyond compliance by designing for reuse, setting up buy-back schemes and publicly reporting recycled-content targets and progress.

Overall Conclusion

The case studies show that effective e-waste management needs a combination of strong legislation, producer responsibility, convenient collection, technically capable recyclers and public participation. Japan and Switzerland prove that well-designed systems can achieve high collection rates. India has a modern legal framework in the 2022 Rules, but its outcomes depend on enforcement and on including the informal sector, as seen in the Pune SWaCH, Kerala and Dharavi examples. Corporate initiatives such as Dell's show that circular design can be commercially viable.

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